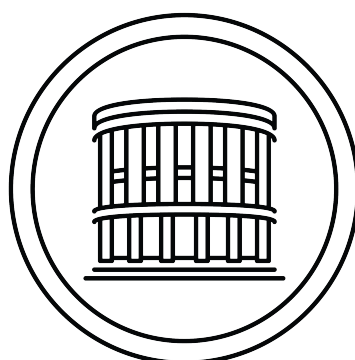


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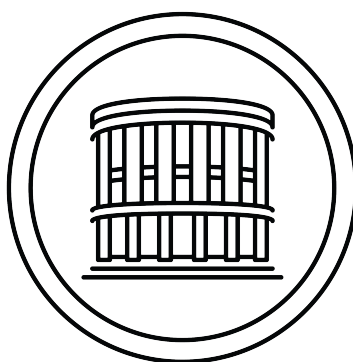


# CONSTRUCTIONS OF HAMILTONIAN GRAPHS WITH PRESCRIBED PARAMETERS

Master thesis



COMENIUS UNIVERSITY IN BRATISLAVA  
FACULTY OF MATHEMATICS PHYSICS AND INFORMATICS



# CONSTRUCTIONS OF HAMILTONIAN GRAPHS WITH PRESCRIBED PARAMETERS

Master thesis

Study program: Applied Informatics  
Branch of study: Applied Informatics  
Department: Department of Applied Informatics  
Supervisor: doc. RNDr. Tatiana Jajcayová, PhD





## THESIS ASSIGNMENT

**Name and Surname:** Bc. Jakub Šmihul'a  
**Study programme:** Applied Computer Science (Single degree study, master II. deg., full time form)  
**Field of Study:** Computer Science  
**Type of Thesis:** Diploma Thesis  
**Language of Thesis:** English  
**Secondary language:** Slovak

**Title:** Constructions of Hamiltonian graphs with prescribed parameters

**Annotation:** Constructions of cages and record graphs, as well as study of their properties, is a very active area of research in extremal graph theory. For each pair of parameters  $(k, g)$  there exists infinitely many  $(k, g)$ -graphs. The graph of the smallest order,  $n(k, g)$ , among these graphs is called a  $(k, g)$ -cage. Cages are known only for few pairs of parameters. It is computationally very difficult to prove that some  $(k, g)$ -graph is a cage. The smallest known  $(k, g)$ -graph, that was not yet proven to be a cage is called a record  $\text{rec}(k, g)$ -graph. Sachs in his work conjectured that for each pair of parameters  $k, g$ , there exists a cage that is Hamiltonian. Therefore, investigating the Hamiltonicity of cages and record graphs is interesting.

**Aim:** The aim of the theses is to develop graph constructions that for any given  $(k, g)$ -graph will return hamiltonian  $(k, g)$ -graph. We will be interested in such graphs of smallest possible order, or at least in some upper bounds for orders of such graphs. This will require to create and program clever algorithms for graph constructions, and also run highly non-trivial computer experiments to test graph properties.

**Keywords:** extremal graphs, girth, valency, hamiltonicity

**Supervisor:** doc. RNDr. Tatiana Jajcayová, PhD.  
**Department:** FMFI.KAI - Department of Applied Informatics  
**Head of department:** doc. RNDr. Tatiana Jajcayová, PhD.

**Assigned:** 19.11.2025

**Approved:** 19.11.2025  
prof. RNDr. Roman Ďurikovič, PhD.  
Guarantor of Study Programme

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Student

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Supervisor



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## ZADANIE ZÁVEREČNEJ PRÁCE

**Meno a priezvisko študenta:** Bc. Jakub Šmihul'a  
**Študijný program:** aplikovaná informatika (Jednoodborové štúdium, magisterský II. st., denná forma)  
**Študijný odbor:** informatika  
**Typ záverečnej práce:** diplomová  
**Jazyk záverečnej práce:** anglický  
**Sekundárny jazyk:** slovenský

**Názov:** Constructions of Hamiltonian graphs with prescribed parameters  
*Konštrukcie hamiltonovských grafov s predpísanými parametrami*

**Anotácia:** Konštrukcie kliebok a rekordných grafov, ako aj štúdium ich vlastností, sú veľmi aktívnou oblasťou výskumu v extrémálnej teórii grafov. Pre každú dvojicu parametrov  $(k, g)$  existuje nekonečne veľa  $(k, g)$ -grafov. Graf najmenšieho rádu,  $n(k, g)$ , spomedzi týchto grafov sa nazýva  $(k, g)$ -klietka. Klietky sú známe len pre niektoré parametre. Výpočtovo je veľmi ťažké dokázať, že nejaký  $(k, g)$ -graf je kliebkou. Najmenší známy  $(k, g)$ -graf, u ktorého sa ešte nedokázalo, že je kliebkou, sa nazýva rekordný  $\text{rec}(k, g)$ -graf. Sachs vo svojej práci predpokladal, že pre každý pár parametrov  $k, g$  existuje klietka, ktorá je hamiltonovská. Preto je skúmanie hamiltonovskosti kliebok a rekordných grafov zaujímavé.

**Cieľ:** Cieľom diplomovej práce je navrhnúť konštrukcie grafov, ktoré pre akýkoľvek daný  $(k, g)$ -graf vrátia hamiltonovský  $(k, g)$ -graf. Budeme sa zaujímať o takéto grafy najmenšieho možného rádu, alebo aspoň o niektoré horné ohraničenia pre rády takýchto grafov. To si bude vyžadovať vytvorenie a naprogramovanie komplexných algoritmov pre konštrukcie grafov a tiež testovanie vlastností grafov pomocou netriviálnych počítačových experimentov.

**Kľúčové slová:** extrémálne grafy, obvod, stupeň, hamiltonicita

**Vedúci:** doc. RNDr. Tatiana Jajcayová, PhD.  
**Katedra:** FMFI.KAI - Katedra aplikovanej informatiky  
**Vedúci katedry:** doc. RNDr. Tatiana Jajcayová, PhD.  
**Dátum zadania:** 19.11.2025

**Dátum schválenia:** 19.11.2025

prof. RNDr. Roman Ďurikovič, PhD.  
garant študijného programu

.....  
študent

.....  
vedúci práce

I hereby declare that I have written this thesis by myself, only with help of referenced literature, under the careful supervision of my thesis advisor.

Bratislava, 2026

.....  
Bc. Jakub Smihula

# Acknowledgement

I would like to thank doc. RNDr. Tatiana Jajcayová, PhD for his supervision and guidance. *[TODO: expand acknowledgements.]*



# Abstract

*[TODO: write English abstract.] ....*

**Keywords:** TODO

# Abstrakt

*[TODO: napísať slovenský abstrakt.] ...*

**Kľúčové slová: TODO**

# Contents

|          |                                           |          |
|----------|-------------------------------------------|----------|
| <b>1</b> | <b>Introduction</b>                       | <b>1</b> |
| <b>2</b> | <b>Motivation</b>                         | <b>2</b> |
| <b>3</b> | <b>Preliminaries and Definitions</b>      | <b>3</b> |
| <b>4</b> | <b>Hamilton Cycles</b>                    | <b>4</b> |
| <b>5</b> | <b>Hardware and Software Tools</b>        | <b>5</b> |
| <b>6</b> | <b>General Construction Techniques</b>    | <b>6</b> |
| <b>7</b> | <b>New Constructions and Main Results</b> | <b>7</b> |
| <b>8</b> | <b>Conclusion</b>                         | <b>8</b> |

# List of Figures

# List of Tables

# Chapter 1

## Introduction

*[TODO: write introduction.]*

# Chapter 2

## Motivation

*[TODO: write motivation.]*

# Chapter 3

## Preliminaries and Definitions

*[TODO: write Preliminaries and Definitions]*



# Chapter 4

## Hamilton Cycles

*[TODO: write Hamilton Cycles.]*

# Chapter 5

## Hardware and Software Tools

*[TODO: write Hardware and Software Tools.]*

# Chapter 6

## General Construction Techniques

*[TODO: write General Construction Techniques.]*

# Chapter 7

## New Constructions and Main Results

*[TODO: write New Constructions and Main Results.]*

# Chapter 8

## Conclusion

*[TODO: write Conclusion.]*

# Bibliography

- [1] Geoffrey Exoo and Robert Jajcay. Dynamic cage survey. *The electronic journal of combinatorics*, pages DS16–Jul, 2012.
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